```
// Bluetooth HC-06/HC-05   <----> Arduino SENSor Shield V5.0 (4pin
TX,RX,-,+)
//      RX                 -NOI-            TX
//      TX                 -NOI-            RX
//      GND                -NOI-            -
//      +5V                -NOI-            +

//#include <LiquidCrystal_I2C.h> //Includes I2C-LCD1602 lib
#include <Wire.h> //Includes I2C Lib

#include "UCNEC.h"
int32_t temp = 0;
UCNEC myIR(A0);
#define F 16718055
#define B 16732845
#define L 16713975
#define R 16734885
#define S 16719075

// Motor control pins: L298N H bridge
const int enAPin = 6; // Left motor PWM speed control
const int in1Pin = 7; // Left motor Direction 1
const int in2Pin = 5; // Left motor Direction 2
const int in3Pin = 4; // Right motor Direction 1
const int in4Pin = 2; // Right motor Direction 2
const int enBPin = 3; // Right motor PWM speed control

int command;                //bien luu trang thai Bluetooth
int speedCar    = 160;      // toc do 50 - 255.
int speed_Coeff = 2;        // he so suy giam <=> giam may lan toc do

/************************Define Line Track
pins************************************/
const int SensorLeft   = 10;       //Left sensor input (A5)
const int SensorMiddle = 9;       //Midd sensor input (A4)
const int SensorRight  = 8;       //Right sensor input(A3)
int SL;    //Status of Left line track sensor
int SM;    //Status of Midd line track sensor
int SR;    //Status of Righ line track sensor
void goAhead(int speedcar1, int speedcar2){

      digitalWrite(in1Pin,LOW);
      digitalWrite(in2Pin,HIGH);         // LEFT tien
      digitalWrite(in3Pin,LOW);
      digitalWrite(in4Pin,HIGH);         // RIGHT tien
      analogWrite(enAPin, speedcar1);
      analogWrite(enBPin, speedcar2);

  }

void goBack(int speedcar1, int speedcar2){
      digitalWrite(in1Pin,HIGH);
      digitalWrite(in2Pin,LOW);         // LEFT lui
      digitalWrite(in3Pin,HIGH);
```

```
      digitalWrite(in4Pin,LOW);          // RIGHT lui
      analogWrite(enAPin, speedcar1);
      analogWrite(enBPin, speedcar2);
  }

void goRight(int speedcar1, int speedcar2){
      digitalWrite(in1Pin,LOW);
      digitalWrite(in2Pin,HIGH);          // LEFT tien
      digitalWrite(in3Pin,HIGH);
      digitalWrite(in4Pin,LOW);           // RIGHT lui
      analogWrite(enAPin, speedcar1);
      analogWrite(enBPin, speedcar2);

  }

void goLeft(int speedcar1, int speedcar2){
      digitalWrite(in1Pin,HIGH);
      digitalWrite(in2Pin,LOW);          // LEFT lui
      digitalWrite(in3Pin,LOW);
      digitalWrite(in4Pin,HIGH);          // RIGHT tien
      analogWrite(enAPin, speedcar1);
      analogWrite(enBPin, speedcar2);

  }

void stopRobot(){
      digitalWrite(in1Pin,LOW);
      digitalWrite(in2Pin,LOW);
      digitalWrite(in3Pin,LOW);
      digitalWrite(in4Pin,LOW);
      analogWrite(enAPin, 0);
      analogWrite(enBPin, 0);
}
void setup() {

      pinMode(enAPin, OUTPUT);
      pinMode(in1Pin, OUTPUT);
      pinMode(in2Pin, OUTPUT);
      pinMode(in3Pin, OUTPUT);
      pinMode(in4Pin, OUTPUT);
      pinMode(enBPin, OUTPUT);

      Serial.begin(9600); //Init Serial port rate set to 9600

       pinMode(SensorLeft,  INPUT);    //Init left sensor
       pinMode(SensorMiddle, INPUT);   //Init Middle sensor
       pinMode(SensorRight, INPUT);    //Init Right sensor
}

void loop() {
                SL = digitalRead(SensorLeft);
                SM = digitalRead(SensorMiddle);
                SR = digitalRead(SensorRight);
```

```
        //right fast 20 200
        if (SM == HIGH && SR == HIGH && SL == LOW)
        {
              goAhead(160,40);delay(100);
        }
        if (SM == HIGH && SR == LOW && SL == HIGH)
        {
              goAhead(40,160);delay(100);

        }
        if (SM == HIGH && SR == LOW && SL == LOW)
        {
              goAhead(160,160);
        }
        if(SM == LOW && SR == LOW && SL == HIGH)
        {
              goAhead(80,200);delay(100);
        }
        if(SM == LOW && SR == HIGH && SL == LOW)
        {
              goAhead(200,80);delay(100);
        }
        else
        {
              stopRobot(); delay(100);
        }

}
```